Testes

Testes are the reproductive glands of the males. They are composed of seminiferous tubules, Leydig cells and Sertoli cells. **Leydig cells** form the endocrine part of the testes. They secrete the male sex hormone called **testosterone**.

Functions of testosterone

- It influences the process of spermatogenesis.
- It stimulates protein synthesis and controls muscular growth.
- It is responsible for the development of secondary sexual characters (distribution of hair on body and face, deep voice pattern, etc).

Ovary

The ovaries are the female gonads located in the pelvic cavity of the abdomen. They secrete the female sex hormones

a. Estrogen

b. Progesterone

Estrogen is produced by the **Graafian follicles** of the ovary and **progesterone** from the **corpus luteum** that is formed in the ovary from the ruptured follicle during ovulation.

Functions of estrogens

- It brings about the changes that occur during puberty.
- It initiates the process of oogenesis.
- It stimulates the maturation of ovarian follicles in the ovary.
- It promotes the development of secondary sexual characters (breast development, high pitched voice etc).

Functions of progesterone

- It is responsible for the premenstrual changes of the uterus.
- It prepares the uterus for the implantation of the embryo.
- It maintains pregnancy.
- It is essential for the formation of placenta.

16.2.7 Thymus Gland

Thymus is partly an endocrine gland and partly a lymphoid gland. It is located in the upper part of the chest covering the lower end of trachea. **Thymosin** is the hormone secreted by thymus.

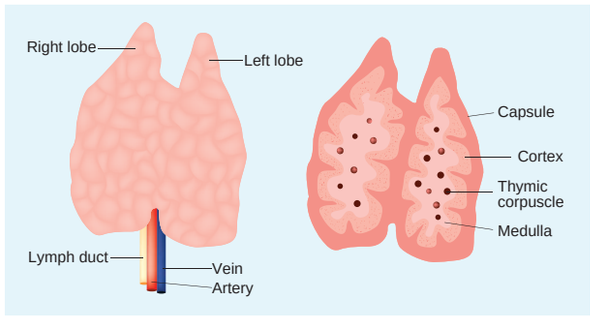


Figure 16.12 Thymus Gland

Functions of Thymosin

- It has a stimulatory effect on the immune function.
- It stimulates the production and differentiation of lymphocytes.

Points to Remember

- Auxins are produced at the tip of stems and roots from where they migrate to the zone of elongation.
- Cytokinins are the plant hormones that promote cell division or cytokinesis in plant cells.
- Gibberellins induce the formation of seedless fruit and parthenocarpic fruits.
- Abscisic acid is a growth inhibitor which regulates abscission and dormancy. It increases tolerance of plants to various kinds of stress.
- Ethylene is a gaseous plant hormone mainly concerned with maturation and ripening of fruits.

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